

CareerNova: AI-Powered Career Guidance and Placement Intelligence Platform

A PROJECT REPORT

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DECLARATION

We hereby declare that the work presented in the report entitled CAREERNOVA: AI-Powered Career Guidance and Placement Intelligence Platform is an authentic record of our own work carried out during the period from January 2026 to April 2026 at the School of Computer Science Engineering and Technology, Bennett University, Greater Noida. The matter and results presented in this report have not been submitted by us for the award of any other degree elsewhere.

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ABSTRACT

Undergraduate career planning is usually characterized by inadequate guidance role-specific preparations, and less data-driven feedback. The generic advice that is normally given to students fails to consider the students branch, strengths, interests, learning patterns, placement preparedness or the current market demand. To overcome this hindrance, we designed CareerNova, a Web based AI solution that offers individualized career counseling, placement preparedness forecasting, skill-gap suggestions, roadmap creation, resume ATS analysis, and industry specific question and answer help via a single assistant.

CareerNova is developed as a full-stack application based on Flask as a backend service SQLite as persistent storage bootstrap as the responsive frontend design and machine learning modules based on scikit-learn. The analytical core is composed of two supervised learning pipelines. The former forecasts readiness to placement using performance and academic measures. The second one conducts career fit and skill-gap analysis using multi-dimensional profile information: interests, strengths, personality tendencies, and goals. The system also has a rules-and-retrieval assistant called Nova AI to offer project-aware and module-aware guidance so it can allow explainable user interaction and actionable guidance.

The site allows safe authentication, history of reports, export of PDFs, management of market trends by the administrator, tracking of tasks, and monitoring of resume versions. To train and test predictive models in cases when real institutional datasets were not available, synthetic yet domain-consistent datasets were created. The system was subjected to route-level and workflow-level checks to make sure that there is consistency and reliability among modules.

The provided solution proves that a unified AI and software engineering solution can enhance student decision support greatly by integrating prediction, explanation, and execution planning in one platform. CareerNova offers a viable and scalable base to institution-level implementation with future possibilities of real-time labor market APIs, LLM integration, and intelligent recommendation learning based on student outcomes.

INTRODUCTION

The contemporary placement system is highly competitive and students are expected to multitask and have a balance between academics, knowledge on code, aptitude training, knowledge on communication, depth of project and market awareness. Nevertheless, a cohesive system that would transform their present profile into precise and quantifiable subsequent steps is lacking in the majority of the students. The conventional approaches to mentoring tend to be manual, slow and inapplicable to high numbers of students. Consequently, there is a tendency among students to make career choices according to trends and not based on a fit, and often there are gaps found out too late into their final year.

Rather than recommending domains of careers alone, the system can gauge readiness, highlight weak areas, and provide viable avenues to enhance placement results. The project integrates machine learning forecasts, profile analytics with guided action planning into a single workflow-driven web application. The feature of tracking and explanation of historical reports also enhances transparency and trust of users in AI-based recommendations.

The project is in line with the present trends in education technology, human-centred AI, and employability analytics. It is made to be accessible, extensible, and implemented in capstone environments in practice.

1.1 Problem Statement

The following are the problems encountered by students who are preparing to do placements:

Absence of personal and elucidable career counseling.

Failure to measure existing readiness to placement.

No systematic skill gaps in view of target roles.

Low fit between the preparation of students and the market.

Lack of unified tracking on resume quality, tasks and progress. Therefore, an all-in-one AI-powered framework capable of analyzing each student profile, creating individual recommendations, and turning recommendations into practical plans with quantifiable results are in demand.

BACKGROUND RESEARCH

Career guidance systems nowadays have evolved into a system of data driven recommenders and predictive employability models as compared to the previous systems of counseling which are quite static. First systems were manual and based on manually maintained taxonomies. Although interpretable, they were not flexible and included personalized ranking systems.

The studies conducted in the data mining of education show that academic background, behavioral indicators, and ability characteristics can predict student performance. Random Forest, Logistic Regression, and Gradient Boosting are commonly applied in the employability prediction due to their ability to deal with heterogeneous characteristics and non-linear correlations. Explainability is crucial in student situations; consequently, hybrid designs, which integrate model outputs and domain rules, tend to be desirable.

Student preparation systems also affect by resume screening in hiring pipelines. Applicant Tracking System (ATS) style filtering is in most cases based on the structure of the documents, keywords of the roles and summarized evidence of impact. ATS-aware feedback tools have been found to be beneficial to early-stage applicants.

Guides are even more accessible through conversational interfaces. Domain-grounded assistants may decrease the hesitation of students by giving them immediate answers, explaining recommendations, and transforming general objectives into weekly specific actions.

CareerNova follows a hybrid strategy basing on this background:

Supervised ML on prediction.

Explainability and practical guidance rule-based logic.

External alignment market trend information.

Formal workflow design in the form of continuous improvement.

2.1 Proposed System

The proposed CareerNova is a web platform in modules, enabling students to:

Make profile and securely log-in.

Assess placement preparedness using predictive ML.

Examine career-role compatibility and determine deficiencies in skills.

Create individual roadmap and task plan.

Check the quality of resumes with ATS checks.

Talk to a helper to have a contextualized direction.

Export and save reports to be used in academics and placement.

2.2 Goals and Objectives

Table 1: Goals and Objectives.

Give individualized and justifiable career suggestions.

Predict placement readiness by multi-factor student inputs.

Find a gap between the skills and chart to improvement plans.

Include market-consciousness in student decision-making.

Enhance the quality of resumes with ATS-based checks.

Retain history, traceability and exportability of reports.

Ensure responsive and user-friendly interface across devices.

Create a scalable architecture of the future institutional scale.

PROJECT PLANNING

3.1 Project Lifecycle

The team used an iterative lifecycle based on Agile with brief implementation cycles:

Module definition and gathering of requirements.

Basic backend and database system.

Creation of ML dataset and training models.

Development and testing of modules.

Integration, refinement of UX, and validation.

This method allowed incremental delivery, on-going QA and early detection of logic and model problems.

3.2 Project Setup

Table 2: Project Set up Choices.

Programming language: Python 3.x

Backend framework: Flask

Frontend: Bootstrap, JavaScript, CSS, HTML.

Database: SQLite/SQLAlchemy ORM.

ML library: scikit-learn

Versioning: Git workflow in repositories.

Report export: Library of PDF generation.

Test strategy: Route and workflow-level QA scripts.

3.3 Stakeholders

Table 3: Roles and Stakeholders.

Mentor: Technical direction and check.

Faculty evaluator: Quality validation and assessment.

Project team: Design, development, testing, documentation.

End users: Students in need of career and placement assistance.

Admin user: Keeps track of market trends and data at the system level.

3.4 Project Resources

Table 4: Resources

Python environment student development machines.

Flask and ML software requirements.

Training and trend seeding datasets.

Local host server and test browser.

Mentor review meetings every week.

3.5 Assumptions

Table 5: Assumptions

Students will give realistic profile and skill entries.

Artificial data sets estimate the variation in students.

Users have primitive internet and web browsers.

It is acceptable to deploy capstone locally.

Team has enough time to test and do documentation.

PROJECT TRACKING

4.1 Tracking

The execution of the project was monitored by:

Incremental commits and version control on a source level.

QA scripts to check routes and module results.

Post-major feature regression.

UI and report export exploratory tests are done manually.

Table 6: Tracking Artifacts

Storing of the code: Git-based repository.

Tracking issues: Team log and sprint checklist.

Regression test: Auto QA test.

Build test: Python compile tests

Documentation: Draft project report files.

4.2 Communication Plan

Table 7: Regular Meetings

Internal group check-in: Weekly.

Mentor review: Weekly

Milestone assessment: At the end of every major module.

Last preparation check: Pre-submission.

Table 8: Internal Exchange of Information.

Status and ownership of modules.

Bug reports and test results.

Updates of the datasets and models.

Documentation progress

Table 9: Information to External Stakeholders

Weekly progress summary

Preview snapshots and important features.

Mitigation and risks updates.

Last deployment and report package

4.3 Deliverables

Table 10: Deliverables

Full-stack CareerNova web application

Synthetic datasets and trained ML models.

Final technical report

Artifacts of presentation and demonstration.

SYSTEM ANALYSIS AND DESIGN

5.1 Overall Description

CareerNova follows a layered architecture with presentation, application logic, model inference, and persistence components. The frontend provides module-specific forms and dashboards through responsive templates. User requests are processed by Flask routes, where input validation, scoring logic, ML inference, and report composition are executed. Database models store user identities, profiles, generated reports, tasks, monthly readiness, and resume versions. The system supports both authenticated user workflows and admin workflows.

The machine learning layer is integrated as persisted model artifacts loaded at runtime. Placement and career-gap prediction modules convert form inputs into feature vectors and generate probability-informed outputs. To improve practical usefulness, model outputs are blended with domain logic and recommendation rules. This hybrid design produces readable recommendations while maintaining predictive structure.

The platform includes a guided assistant interface for fast query handling. Instead of generic responses only, the assistant supports project-aware and module-aware Q and A, helping users understand predictions and choose next actions.

5.2 Users and Roles

Table 11: Users and Roles

Student user

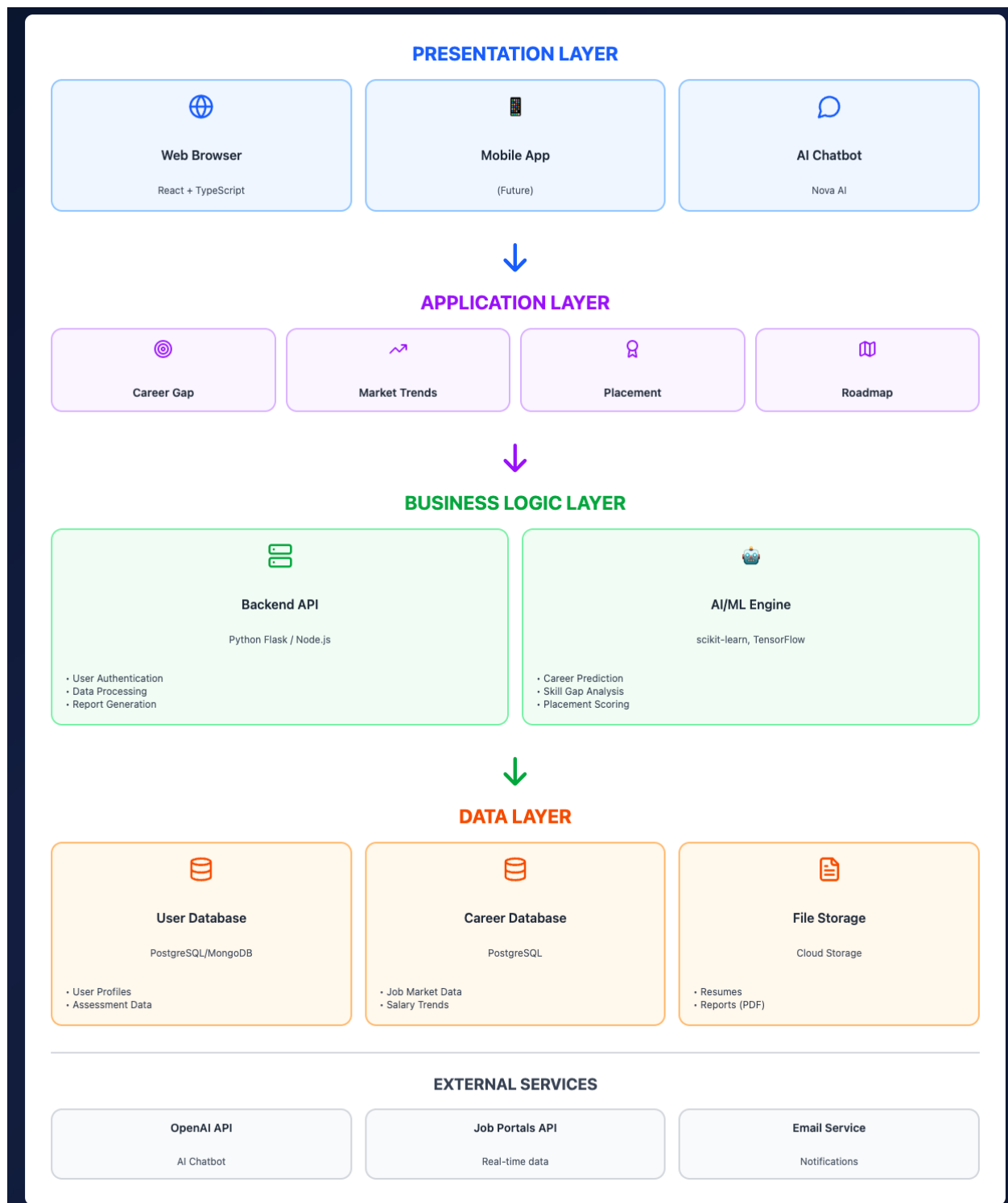
Description: Registers, completes profile, runs modules, tracks progress, and exports reports.

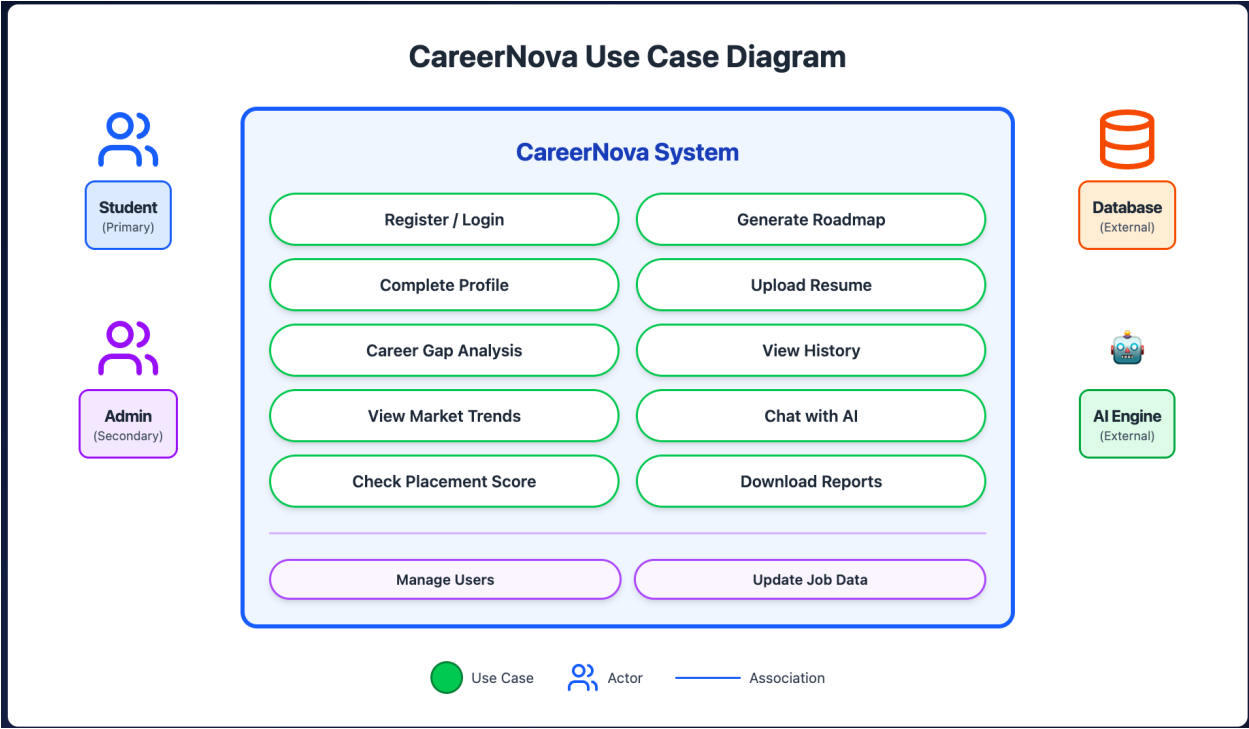
Admin user

Description: Maintains market trend entries and verifies data quality.

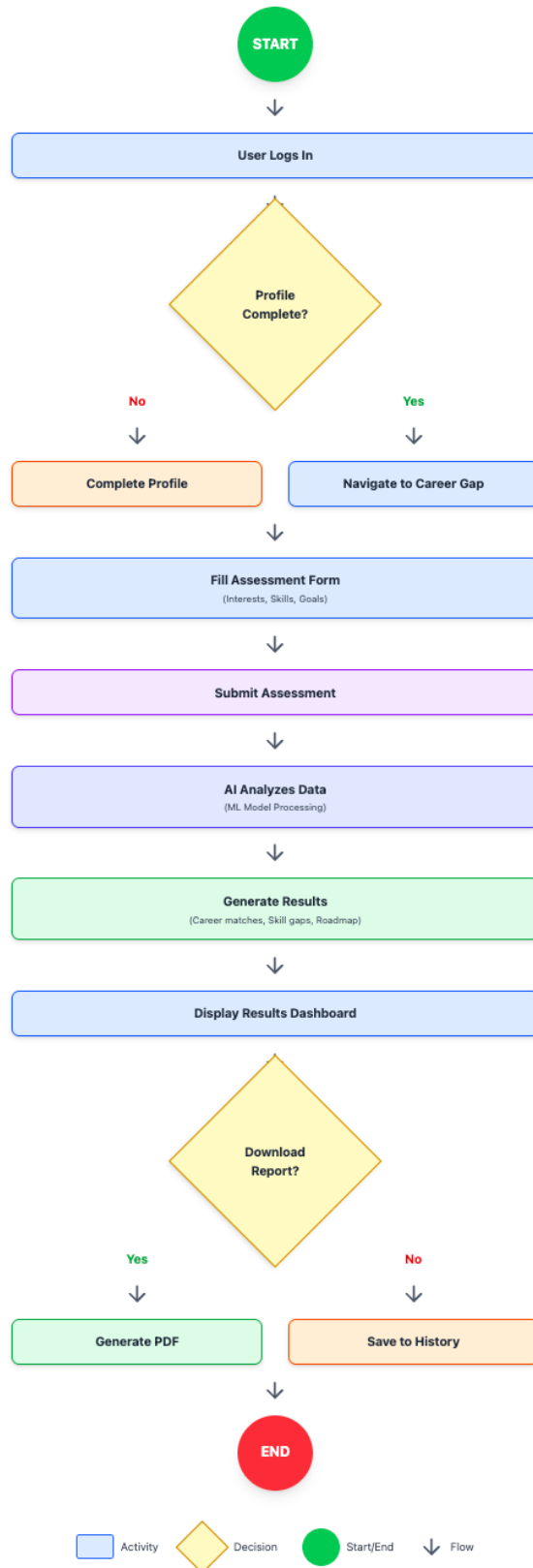
Mentor or evaluator

Description: Reviews outputs, validates workflow quality, and assesses project utility.





Career Gap Analysis - Activity Flow



USER INTERFACE

6.1 UI Description

The interface is student-friendly and has a low cognitive load:

Module-card dashboard-based navigation.

Form inputs sorted by concept and difficulty.

Progress-oriented result panels with suggestions.

Laptop/mobile responsive behavior.

Theme switch accessibility options.

Floating doubt solve button.

6.2 UI Mockup Guidance

Include screenshots for:

Module cards and home page hero.

Career Gap Analyzer form and result panel.

Result view of Placement Predictor.

Weekly action plan.

Resume Tracker and ATS score card.

Interaction of assistant chat panel.

Admin market trend dashboard.

ALGORITHMS AND PSEUDO CODE OF CORE FUNCTIONALITY

Input: Student profile and performance measurements.

Output: Placement readiness score and advice.

Pseudo code:

Read branch aptitude communication projects. certifications coding activity.

Normalise and encode input features.

Load trained placement model.

Calculate predictproba of readiness class.

Combine probability of models with rule-based preparative elements.

Create category label and customized tips.

Save report.

Algorithms of Recommendation

7.2 Career Gap Recommendation.

Input: Interests, strengths, goals, personality, academics.

Output: Preferred career choices and skill gaps.

Pseudo code:

Profile Transform to feature vector.

Compute class probabilities for career roles.

Calculate domain mapping rule-based fit score.

Combine ML probability and rule score using weighted formula.

Rank careers by overall score.

To get the best careers, obtain list of missing skills.

Save suggestions and description.

7.3 Resume ATS Scoring Algorithm

Input: Resume text

Output: ATS score and blank sections.

Pseudo code:

Read PDF or text document.

Check presence of vital areas: skills, projects, education.

Keywords coverage.

Implement formatting and quality checks.

Calculate weighted score and grammar penalty.

Return score and missing components.

7.4 Nova Assistant Retrieval Algorithm

Input: User question

Output: Contextual answer

Pseudo code:

Change message to lower-case.

Assess project knowledge base key word and score of fuzzy match.

Score above threshold, project-aware response.

In case there is a question to determine why recommendation, retrieve most recent report and clarify.

Fallback guidance on return not a confident match.

PROJECT CLOSURE

8.1 Goals and Vision Achievement

The initial concept was to create a viable student guidance platform, which integrates prediction and practical planning. This was accomplished through the provision of a multi-module, full-stack platform through which users can evaluate the status quo, comprehend gaps and trace improvements. The project went further than the one-time recommendation and incorporated explainability, report persistence, and guide by an assistant.

8.2 Delivered Solution

Delivered components include:

User management and authentication.

Career Gap Analyzer (ML) and rule blending.

Ready Predictor with readiness score.

Market Analysis and role-demand analysis.

Learning Roadmap generation.

ATS feedback Resume Tracker.

PDF exports of history management.

Administrative panel to update trends.

Nova AI project and module Q and A assistant.

Artificial data, trained models, and Quality assurance tests.

assess actual placement outcomes data to better calibrate the model.

Add interpretable AI images like importance of features per prediction.

Add voice assistant and multilingual.

Deploy using cloud infrastructure and role based access hardening.

Integrate LLM fallback for more open-ended natural conversation.

Introduce adaptive learning plans using longitudinal student performance

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